



GOA UNIVERSITY

B.E - (Computer) (Sem-VII)(Revised Course 2019-2020)

Paper Title: Neural Networks and Deep Learning (Honours /Minor)

Paper Code: CEAM 04

[Duration: 3 Hours]

[Total Marks: 100]

- Instructions:
- 1) Answer any 5 questions in all.
 - 2) Answer any 2 questions each from Part-A and Part-B
 - 3) One question has to be answered from Part-C
 - 4) Figures to the right indicate marks.

PART A

- Q1 a) Explain the structure and working of the artificial neuron model. Also give a description of any commonly used single neuron model. 06
- b) Describe the Sigmoid activation function. Emphasize on its use in neural networks. 06
- c) Implement AND function using the Mc Culloch Pitts neuron model. 08
- Q2 a) Discuss the difference between cumulative weight adjustment and incremental updating. 06
- b) Write a short note on the concept of generalization in a model. 06
- c) List and elaborate any four performance metrics used to evaluate models. 08
- Q3 a) Discuss the effect of sample size on a neural network. 06
- b) Elaborate on the concept of a feed forward network. 06
- c) Use K means clustering algorithm to divide the following data into 2 clusters. 08

X1	1	2	2	3	4	5
X2	1	1	3	2	3	5

PART B

- Q4 a) What is batch normalization? Why is it required? 06
b) Explain the vanishing gradient problem. 06
c) List and explain the different types of layers found in a Convolutional Neural Network. 08
- Q5 a) Explain the working principle of recurrent neural networks. 06
b) What is the long short term memory concept? What kind of applications is it suited for? 06
c) Write a short note on deep belief networks. 08
- Q6 a) What is the use of loss functions in deep learning? 06
b) What is AlexNet used for? Explain the architecture of this model. 08
c) Explain the concept of long term dependencies. 06

PART C

- Q7 a) What is competitive learning? How is it different from the traditional approach? 06
b) Explain the back propagation algorithm. 08
c) What are self-organizing maps? Write the algorithm for the same. 06
- Q8 a) Explain the convolution operation. 06
b) Write a short note on the various applications of deep learning neural networks. 06
c) Explain the architecture of generative adversarial networks. 08